

Does the Weather Still Move Markets?

A Pre-Registered, Two-Hemisphere, Post-Publication Test

Jason Li*

Working paper — September 28, 2026

Abstract

We re-examine weather and seasonal-mood effects in stock returns with two pre-registered studies, free and replicable data, and a placebo-city inference procedure that directly addresses the concern that weather effects are data-mined. Study 1 uses daily US (1955–2026) and Australian (1984–2026) returns. After replicating [Hirshleifer and Shumway \(2003\)](#) and [Kamstra et al. \(2003\)](#) on their own samples, one of nine primary tests survives multiple-testing correction and placebo inference: New York cloud cover lowers same-day US returns by about 2 basis points per standard deviation (Holm $p = 0.029$), more than the weather of any of 100 placebo cities. Study 2, pre-registered after Study 1, pools 26 further exchanges and seven weather variables. Rain days lower returns (1.6 basis points; Holm $p = 0.045$, beyond all placebo draws), and high air pressure and sunshine point the same way. However, the international cloud effect is absent after the [Hirshleifer and Shumway \(2003\)](#) sample ends, and every fair-weather effect is concentrated before 2003. Seasonal affective disorder effects do not appear in the southern hemisphere, and temperature does not move markets. Weather effects were real but small, and have largely faded since publication; they are not exploitable after trading costs.

Keywords: weather, investor mood, seasonal affective disorder, pre-registration, replication, placebo inference

JEL: G12, G14, G41

*E-mail: lij62766@gmail.com. The pre-registrations, data pipeline and all code are available at [repository URL]. Each pre-registration was committed to version control before the results it covers were estimated.

1 Introduction

Few findings in behavioural finance are as memorable, or as contested, as the claim that sunshine lifts stock prices. [Saunders \(1993\)](#) reported that New York Stock Exchange returns were lower on cloudy days in New York; [Hirshleifer and Shumway \(2003\)](#) extended the result to 26 exchanges; [Kamstra et al. \(2003\)](#) argued that seasonal affective disorder (SAD) produces a seasonal cycle in returns that reverses between hemispheres; and [Cao and Wei \(2005\)](#) linked low temperatures to higher returns. Each finding was challenged soon after publication. [Trombley \(1997\)](#) showed that the New York effect was sensitive to the sample and the definition of cloudiness; [Jacobsen and Marquering \(2008\)](#) argued that SAD and temperature effects are hard to distinguish from the Halloween effect of [Bouman and Jacobsen \(2002\)](#); [Kelly and Meschke \(2010\)](#) questioned the construction of the SAD variable; and [Novy-Marx \(2014\)](#) showed that a researcher free to choose predictors can find “significant” relations with sunspots and planetary alignments. For Australia, the evidence has always been weak ([Keef and Roush, 2007](#); [Worthington, 2009](#)).

Two decades later the question can be asked in a way that was not possible when these papers were written. More than twenty years of data have accumulated since publication, which gives a genuine out-of-sample test in the spirit of [McLean and Pontiff \(2016\)](#). Pre-registration makes it possible to commit to specifications before seeing the new data. And free, long, hourly weather records for any city on Earth make it possible to ask how unusual the New York result is relative to the weather in cities that have nothing to do with the US market.

This paper makes four contributions. First, it pre-registers nine primary tests covering cloud cover, temperature and SAD in the US and Australia, their post-publication persistence, a market-structure channel and a small-stock channel, together with a decision rule that requires both a multiple-testing-adjusted p -value and a placebo percentile. Second, it introduces a placebo-city test: the same regression is estimated with weather from 100 cities outside the US and Australia, and the real city’s coefficient is ranked within that distribution. Because the placebo cities share the seasonal structure of weather but not any plausible link to the market, the test speaks directly to the data-mining critique. Third, it uses the reversal of seasons between hemispheres to separate a seasonal-mood channel from calendar effects. Fourth, a second study, pre-registered after the first study’s results were known and run on 26 exchanges whose returns had not been examined, tests seven weather variables (cloud, sunshine, temperature, wind, rain, humidity and pressure) and gives an out-of-sample test of the international results of [Hirshleifer and Shumway \(2003\)](#).

The results are mixed in an informative way. The New York cloud effect survives: -2.1 basis points (bp) per standard deviation of cloud cover over 1955–2026, significant after Holm and Romano–Wolf corrections, and more extreme than every one of 100 placebo cities. But the effect is small, not clearly present after 2003, not visible in Australian data after 2003, and not exploitable after costs. SAD does not appear in Australia, and temperature does not move the market, although warm days lower the returns of energy and utility stocks, consistent with a physical demand channel rather than mood. Across 26 further exchanges, rain days lower returns and high pressure and sunshine raise them, with rain surviving the full decision rule; but these effects are concentrated before 2003, and the international cloud effect is absent after 1997.

2 Hypotheses

Table 2 lists the nine primary tests; the full pre-registration, including the power analysis and decision rule, is in the online repository. All tests are one-sided in the direction the original literature predicts.

H1 Deseasonalised cloud cover in the exchange city lowers same-day index returns (New York for the US, Sydney for Australia).

H2 The cloud effect survives publication: its coefficient after July 2003 is negative.

H3 In Australia, the SAD effect follows the local (southern-hemisphere) season rather than the New York calendar.

H4 A temperature anomaly lowers returns (Cao and Wei, 2005).

H5 The New York weather effect weakens after the NYSE Hybrid Market (January 2007) moved most trading off the floor.

H6 Cloud effects are stronger in small stocks, whose investors are more local and more retail (Loughran and Schultz, 2004).

A primary hypothesis is *supported* only if its one-sided HAC p -value survives a Holm correction at family-wise $\alpha = 0.05$ and the placebo-city percentile lies in the predicted 5% tail; *inconclusive* if only one holds; and *not supported* otherwise. With daily return volatility near 1% and about 5,800 post-2003 days, the minimum detectable effect at 80% power is roughly 3.3 bp per standard deviation, about the size of the original estimates. A post-publication null is therefore informative only through its confidence interval, which we report throughout.

3 Data

Returns. US returns are the daily Center for Research in Security Prices (CRSP) value-weighted market return (market excess return plus the risk-free rate) from Kenneth French’s data library; size and industry portfolios come from the same source. Australian returns are the S&P/ASX All Ordinaries index from 1984, with dividends added back as the average daily dividend return in each calendar month, estimated from the S&P/ASX 200 total-return and price indices (2019–2026). Results with price-only returns are almost identical. All returns are log returns in percent. The sample ends on 31 August 2026.

Weather. Hourly observations come from NOAA’s Global Historical Climatology Network hourly data set for New York LaGuardia (with JFK filling gaps) and from NOAA’s Integrated Surface Database and the Iowa Environmental Mesonet METAR archive for Sydney Airport. Cloud cover is total sky cover in oktas, and temperature is air temperature; both are averaged over 06:00–16:00 local time, following Hirshleifer and Shumway (2003). Each variable is deseasonalised against a trailing 30-year climatology for the same ISO week, using only earlier years, and standardised. The

resulting anomalies are available from 1955 in New York and 1984 in Sydney. For placebo cities, alternative cities and a single-source robustness check we use the ERA5 reanalysis (Hersbach et al., 2020) from the Copernicus Climate Data Store, processed identically. ERA5 cloud cover correlates 0.85 with the LaGuardia station series at the daily frequency.

Seasonal variables. Night length is computed astronomically for each exchange city’s latitude. SAD_t equals night length minus 12 hours during the local autumn and winter, and zero otherwise; $FALL_t$ is a local-autumn dummy (Kamstra et al., 2003).

4 Method

Each test is an OLS regression of the daily return on the weather or seasonal variable and a fixed set of controls: day-of-week dummies, turn-of-year, pre- and post-holiday, and tax-year-end dummies (December in the US, June in Australia), and two lags of the dependent variable. Australian regressions also control for the most recent completed US return, because the New York session closes before Sydney opens. Standard errors are Newey–West with five lags (Newey and West, 1987). Multiplicity across the nine tests is handled with the Holm procedure (Holm, 1979), with a Romano–Wolf stepdown (Romano and Wolf, 2005) based on a joint stationary bootstrap (Politis and Romano, 1994) as robustness.

The placebo-city test re-estimates each weather regression 100 times, each time using ERA5 cloud or temperature anomalies for a different city more than 1,000 km from both New York and Sydney (the list was fixed in the pre-registration). The real city is re-estimated with ERA5 data as well, so the comparison is like for like, and we report the share of placebo coefficients at least as extreme. A second placebo shifts the real station’s weather series by $k = 1, \dots, 30$ years. This preserves seasonality exactly and breaks only the day-to-day link.

5 Results

5.1 Replication

Table 1 shows that our data reproduce the original findings on the original samples. In a logit of up days on deseasonalised New York cloud cover over 1982–1997, the coefficient is -0.114 ($t = -3.56$), matching the sign and significance of Hirshleifer and Shumway (2003). Over 1928–2000, the SAD coefficient is positive (2.2 bp per hour of night beyond 12, $t = 2.77$) and the autumn dummy negative (-5.0 bp, $t = -2.31$), as in Kamstra et al. (2003). Any failure of the extended tests therefore cannot be blamed on data that cannot reproduce the originals.

5.2 Primary tests

Table 2 reports the nine pre-registered tests. Only H1 in the US is supported. A one-standard-deviation increase in New York cloud cover lowers the same-day market return by 2.1 bp (95% CI $[-3.7, -0.6]$). The Holm-adjusted p -value is 0.029 and the Romano–Wolf p -value 0.035. Figure 1

Table 1: Replication gate (data up to 2000). Pass = predicted sign with one-sided $p < 0.10$.

Specification	Pred.	β (bp)	t	p	N
HS 2003 NYC logit (1982-97)	<0	-11.41	-3.56	<0.001	4,012
HS 2003 NYC OLS (1982-97)	<0	-2.01	-1.26	0.105	4,012
KKL 2003 US SAD (1928-2000)	>0	+2.15	+2.77	0.003	19,411
KKL 2003 US FALL (1928-2000)	<0	-5.01	-2.31	0.011	19,411
KKL 2003 US SAD + weather (1955-2000)	>0	+2.63	+3.00	0.001	9,389
KKL 2003 US FALL + weather (1955-2000)	<0	-3.34	-1.46	0.072	9,389

shows the placebo distribution: centred almost exactly on zero (mean -0.03 bp, standard deviation 0.66 bp), with the New York ERA5 coefficient (-1.9 bp) beyond every placebo city. None of the 30 year-shifted series produces a coefficient as negative either.

In Australia the Sydney cloud coefficient has the same sign and similar size (-1.8 bp) and lies in the 2% tail of its placebo distribution, but it does not survive the Holm correction, so H1-AU is inconclusive.

Table 2: Pre-registered primary tests. β in basis points per one-standard-deviation weather shock (for H3, per hour of night beyond 12). One-sided HAC p -values in the predicted direction; Holm and Romano–Wolf adjust across the nine tests. Placebo = share of 100 placebo cities with a coefficient at least as extreme (ERA5); Shift = same for 30 year-shifted weather series.

ID	Hypothesis	β	95% CI	t	p_1	Holm	R–W	Placebo	Shift	Verdict
H1-US	NYC cloud lowers US returns	-2.13	[-3.7, -0.6]	-2.72	0.003	0.029	0.035	0.00	0.00	Supported
H1-AU	Sydney cloud lowers AU returns	-1.77	[-3.6, +0.1]	-1.87	0.031	0.246	0.224	0.02	0.03	Inconclusive
H2-US	US cloud effect after 2003	-1.95	[-5.2, +1.3]	-1.18	0.119	0.834	0.619	0.17	0.17	Not Supported
H2-AU	AU cloud effect after 2003	+0.07	[-2.3, +2.4]	+0.06	0.523	1.000	0.982	0.50	0.53	Not Supported
H3-AU	AU SAD follows the season	-1.38	[-3.9, +1.2]	-1.06	0.855	1.000	0.982	–	–	Not Supported
H4-US	NYC temperature lowers returns	+0.13	[-1.2, +1.5]	+0.19	0.574	1.000	0.982	0.50	0.60	Not Supported
H4-AU	Sydney temperature lowers returns	+0.67	[-0.9, +2.2]	+0.85	0.803	1.000	0.982	0.87	0.93	Not Supported
H5-US	Effect weakens after NYSE Hybrid	+0.18	[-3.9, +4.3]	+0.09	0.465	1.000	0.982	0.36	0.60	Not Supported
H6-US	Stronger in small minus big	-0.07	[-1.2, +1.1]	-0.11	0.454	1.000	0.982	0.10	0.40	Not Supported

After publication. The post-2003 US coefficient is -2.0 bp, almost identical to the full-sample estimate, but its confidence interval ($[-5.2, +1.3]$) cannot distinguish persistence from disappearance. For Australia the post-2003 coefficient is $+0.1$ bp with a much tighter interval ($[-2.3, +2.4]$). Figure 2 shows trailing ten-year estimates. The New York coefficient is negative in most windows but close to zero for windows ending 2005–2012. The Sydney coefficient drifts from about -3 bp in the 1990s towards zero after 2015.

Seasons, temperature and channels. The Australian SAD coefficient has the wrong sign (-1.4 bp per hour, $t = -1.06$), so the seasonal-mood cycle does not follow the southern-hemisphere season (H3). Temperature anomalies have no detectable effect in either market (H4). There is no evidence that the New York effect weakened when floor trading declined (H5), although the interaction is imprecise, nor that it is concentrated in small stocks (H6).

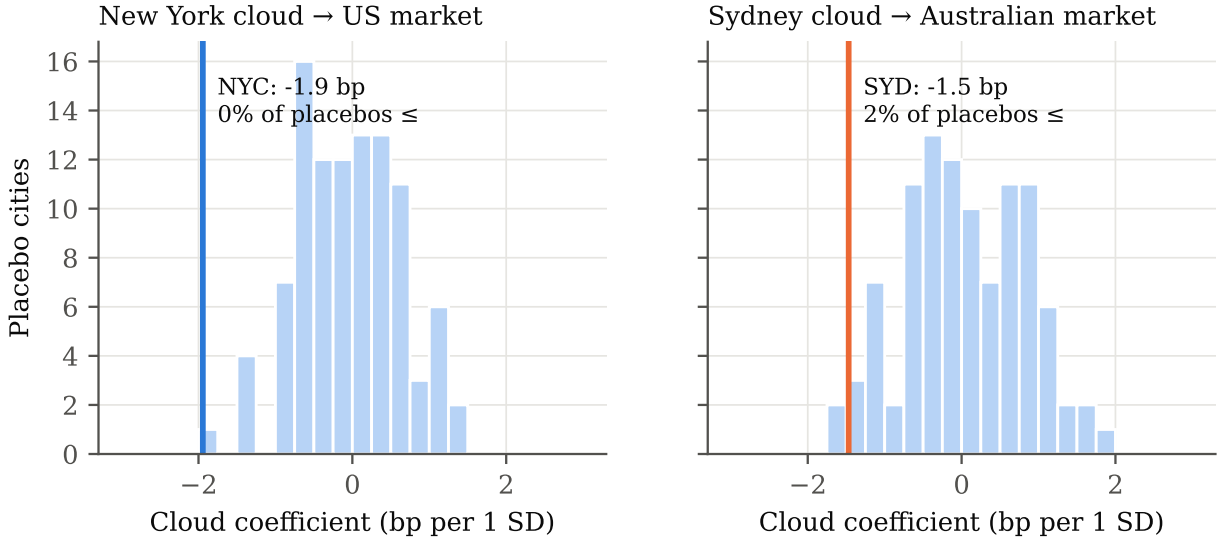


Figure 1: Placebo-city distributions for the cloud coefficient (H1). Bars show coefficients estimated with ERA5 cloud anomalies from 100 placebo cities; the vertical line is the exchange city, also measured with ERA5.

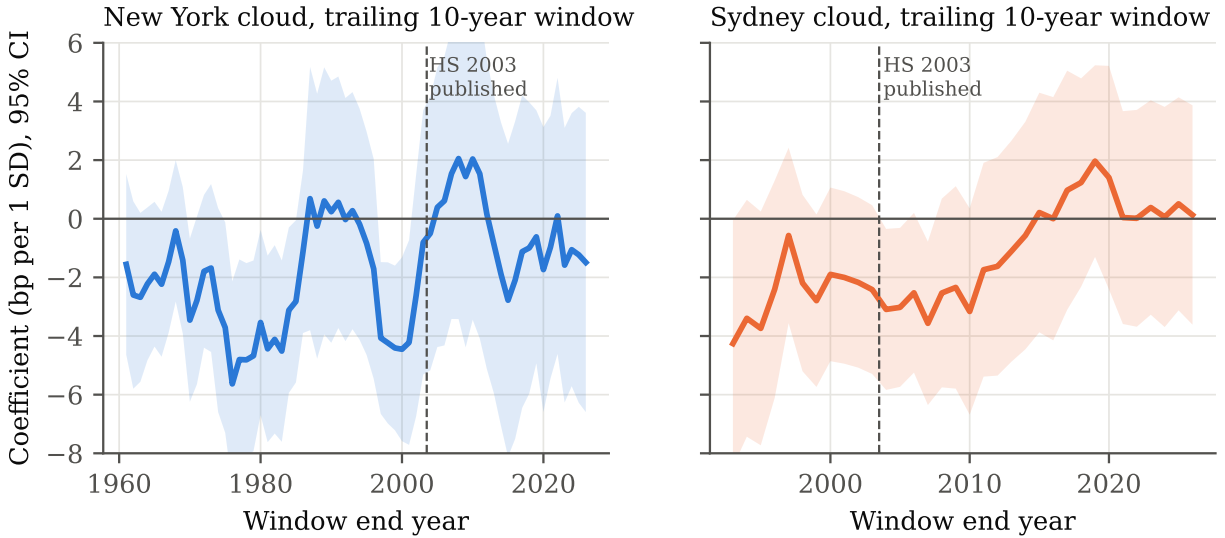


Figure 2: Trailing ten-year estimates of the cloud coefficient, with 95% HAC confidence bands. The dashed line marks the publication of [Hirshleifer and Shumway \(2003\)](#).

5.3 Robustness and exploratory results

Table 6 (appendix) varies the US and Australian cloud and temperature specifications. The US cloud result holds with ERA5 weather (-1.9 bp), full-sample deseasonalisation (-2.1 bp), a logit of up days ($t = -3.73$) and a GARCH(1,1) model with Student- t errors (-2.1 bp, $t = -4.24$). It is weaker with Chicago weather (-1.2 bp, $t = -1.61$), as expected if the relevant weather is where the market is. In a split sample, the coefficient is -2.1 bp ($t = -2.53$) over 1955–2006 and -1.9 bp ($t = -0.97$) over 2007–2026. Rain during trading hours lowers US returns by 5.4 bp ($t = -2.86$), consistent with the cloud result.

The Australian cloud coefficient is robust to price-only returns and ERA5 data, but Melbourne weather shows nothing ($+0.2$ bp). Every post-2003 Australian variant is within 0.6 bp of zero.

Table 7 examines seasonal effects. In the US, the SAD coefficient is positive over 1955–2026 (2.0 bp, $t = 2.40$) and similar but insignificant after 2003. In Australia, neither the local SAD variable (0.2 bp) nor the New York-calendar SAD variable (-0.2 bp) nor a Halloween dummy (1.4 bp, $t = 0.86$) is significant, so the reversed-season test favours neither the mood nor the calendar account. It simply finds no seasonal cycle.

Table 9 provides suggestive evidence for a physical channel. Controlling for the same-day market return, warmer-than-normal days lower the returns of the oil industry (-2.0 bp, $t = -2.87$) and of utilities (-1.0 bp, $t = -2.25$). This is consistent with lower heating demand rather than investor mood. Cloudy days raise coal-industry returns (5.6 bp, $t = 3.82$). These are sixteen exploratory tests, so only the coal and oil results clear a Bonferroni threshold.

5.4 Economic significance

Table 3 asks whether the effect can be traded. Using the SPY exchange-traded fund from 1993, we hold the market from open to close only on days whose early-morning (06:00–10:00) New York cloud cover is below normal. The open-to-close return on sunny mornings exceeds that on cloudy mornings by only 0.3 bp per day ($t = 0.13$). With a 1 bp round-trip cost, every variant loses money. The New York effect, although statistically robust over seven decades, is far too small to trade. This is consistent with the conclusion of [Hirshleifer and Shumway \(2003\)](#) and with its survival as an anomaly.

6 Study 2: Seven weather variables across 26 exchanges

6.1 Design

Study 2 was pre-registered ([prereg/study2_preregistration.md](#)) after Study 1 was complete, and frozen before any return from the new exchanges was examined. It covers every major exchange whose benchmark index has free daily history from 2006 or earlier with fewer than 5% zero-change days: twelve in Europe, nine in Asia, four in the Americas and one in New Zealand. New York and Sydney are excluded from the primary tests because their results were already known. For every city, seven weather variables come from ERA5 and are processed identically: mean cloud cover, solar radiation (“sunshine”), temperature, wind speed, relative humidity and sea-level pressure, each

Table 3: Cloud-timed SPY open-to-close strategy, 1993–2026. Long SPY from the open to the close on days whose 06:00–10:00 New York cloud anomaly is below zero (“sunny”), or above it (“cloudy”); otherwise in cash (zero return). Costs are per round trip.

Rule	Cost (bp)	Ann. return (%)	Sharpe	Days held
every day	0	+0.71	+0.05	8,430
sunny mornings only	0	+0.45	+0.05	3,381
cloudy mornings only	0	+0.25	+0.02	5,049
every day	1	-1.81	-0.12	8,430
sunny mornings only	1	-0.56	-0.06	3,381
cloudy mornings only	1	-1.25	-0.11	5,049
every day	2	-4.33	-0.29	8,430
sunny mornings only	2	-1.57	-0.17	3,381
cloudy mornings only	2	-2.76	-0.23	5,049
Sunny minus cloudy mornings: +0.28 bp per day ($t = 0.13$), $N = 8,430$				

as a trailing-climatology z -score over 06:00–16:00 local time, plus a rain-day indicator (≥ 1 mm) measured relative to its climatological frequency. Returns are scaled by each index’s trailing 60-day volatility, so coefficients are in percent of a daily standard deviation, and estimated in a pooled regression with exchange fixed effects and Driscoll–Kraay standard errors, which allow for correlation across markets on the same day. The decision rule is the same as in Study 1. The placebo test gives each exchange the weather of a random city more than 1,000 km away, 200 times.

6.2 Results

Table 4 reports the nine primary tests. Rain is supported: a rain day lowers returns by 1.6% of a daily standard deviation (about 1.6 bp at the median volatility; Holm $p = 0.045$), and no placebo assignment produces an effect as large. High pressure (+1.0 bp per standard deviation, Holm $p = 0.070$), sunshine (+0.6 bp), cloud (−0.4 bp) and humidity (−0.4 bp) all lie in the extreme tail of their placebo distributions but do not survive the Holm correction, so they are inconclusive. Their signs are consistent: fair weather goes with higher returns. Temperature, wind and SAD show nothing. The SAD test has little power, however, since only four exchanges are in the southern hemisphere.

The out-of-sample cloud test (S2) is clearly negative. Using only data from 1998, after the [Hirshleifer and Shumway \(2003\)](#) sample ends, the pooled cloud coefficient is −0.2 bp ($t = -0.57$) and sits in the middle of its placebo distribution.

Table 5 and Figure 3 show why. Every fair-weather effect is concentrated in the period before July 2003. The rain coefficient falls from −3.8% of a standard deviation ($t = -3.9$) to −0.3% ($t = -0.4$), sunshine from +1.8% ($t = 4.2$) to −0.1%, and cloud from −1.2% ($t = -2.6$) to zero. Pressure declines least, from +1.5% to +0.7% ($t = 1.5$). The effects are found in developed rather than emerging markets and in the northern hemisphere. When all seven variables enter one regression, only pressure remains significant ($t = 2.1$). This suggests that the fair-weather pattern reflects high-pressure weather systems more than any single variable.

Figure 4 shows the exchange-level estimates. Twelve of the 196 exchange–variable tests survive

Table 4: Study 2 pre-registered primary tests, pooled over 26 exchanges not used in Study 1. β is in percent of a daily return standard deviation per one-SD weather shock (rain: per rain day relative to its climatological frequency); bp converts at the median daily volatility. Driscoll–Kraay standard errors. Placebo = share of 200 random placebo-city assignments at least as extreme.

ID	Weather variable	β (% SD)	bp	t	p	Holm	Placebo	N	Verdict
S1	Cloud (full sample)	-0.39	-0.41	-1.49	0.068	0.410	0.005	222,941	Inconclusive
S2	Cloud, 1998 onward	-0.16	-0.17	-0.57	0.284	1.000	0.170	184,234	Not Supported
S3	Sunshine	+0.55	+0.58	+2.14	0.016	0.113	0.010	222,941	Inconclusive
S4	Temperature	-0.12	-0.13	-0.39	0.349	1.000	0.390	222,941	Not Supported
S5	Wind	-0.11	-0.12	-0.41	0.342	1.000	0.550	222,941	Not Supported
S6	Rain day	-1.55	-1.64	-2.58	0.005	0.045	0.000	222,941	Supported
S7	Humidity (two-sided)	-0.42	-0.44	-1.66	0.097	0.484	0.025	222,941	Inconclusive
S8	Pressure (two-sided)	+0.93	+0.98	+2.62	0.009	0.070	0.000	222,941	Inconclusive
S9	SAD follows local season	+0.13	+0.14	+0.22	0.412	1.000	–	222,941	Not Supported

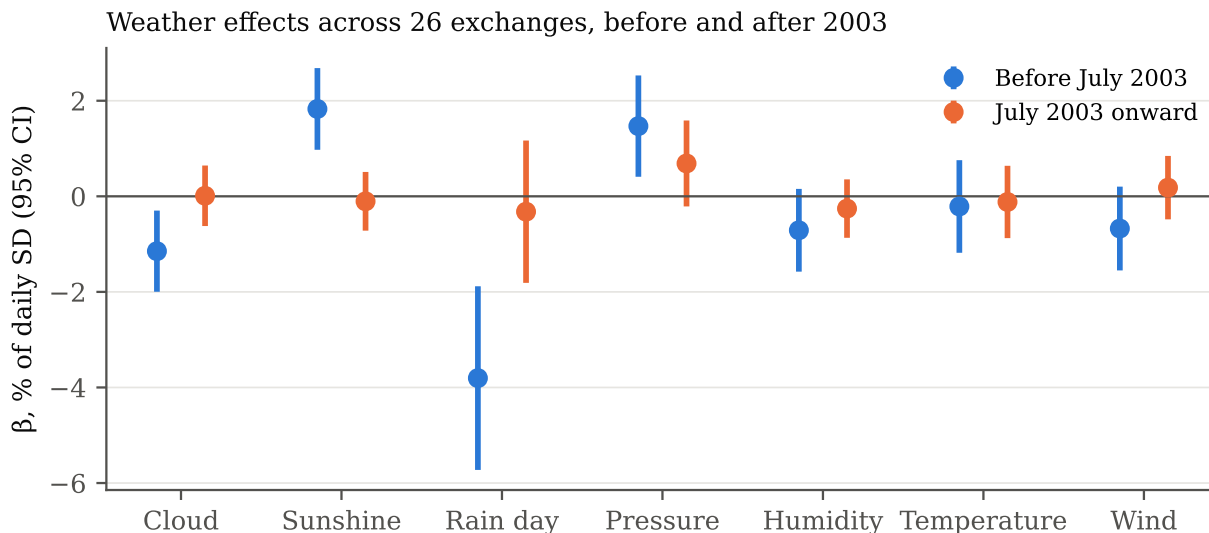


Figure 3: Pooled Study 2 coefficients before and after July 2003, with 95% Driscoll–Kraay confidence intervals.

Table 5: Study 2 exploratory splits (26 exchanges). Cells: β in percent of a daily SD, with t -statistics in parentheses. “Joint” enters all seven anomalies in one regression.

Variable	Pre-2003	Post-2003	Developed	Emerging	Northern	Southern	Joint
Cloud	-1.15 (-2.6)	+0.01 (+0.0)	-0.51 (-1.5)	-0.18 (-0.5)	-0.46 (-1.6)	-0.00 (-0.0)	+0.08 (+0.2)
Sunshine	+1.83 (+4.2)	-0.10 (-0.3)	+0.71 (+2.1)	+0.28 (+0.8)	+0.60 (+2.2)	+0.24 (+0.4)	+0.33 (+0.8)
Temperature	-0.21 (-0.4)	-0.12 (-0.3)	-0.21 (-0.5)	+0.05 (+0.1)	-0.07 (-0.2)	-0.32 (-0.5)	-0.09 (-0.2)
Wind	-0.67 (-1.5)	+0.18 (+0.5)	-0.31 (-0.8)	+0.22 (+0.6)	-0.11 (-0.4)	-0.10 (-0.2)	+0.13 (+0.5)
Rain day	-3.80 (-3.9)	-0.32 (-0.4)	-1.71 (-2.2)	-1.20 (-1.3)	-1.85 (-2.8)	+0.15 (+0.1)	-0.76 (-1.0)
Humidity	-0.71 (-1.6)	-0.26 (-0.8)	-0.53 (-1.5)	-0.24 (-0.7)	-0.46 (-1.6)	-0.26 (-0.4)	+0.02 (+0.0)
Pressure	+1.47 (+2.7)	+0.69 (+1.5)	+1.64 (+3.4)	-0.17 (-0.4)	+1.09 (+2.8)	+0.00 (+0.0)	+0.81 (+2.1)

a Benjamini–Hochberg false-discovery rate of 10%. Pressure effects appear in London, Frankfurt, Paris, Zurich and Brussels, but these cities share weather systems, so they are closer to a single European finding than to five. Tokyo and Taipei show cloud and sunshine effects like New York’s. Among the other exploratory results, storm and heat days have no effect on returns, and warmer-than-normal days are associated with lower volatility ($t = -2.8$).

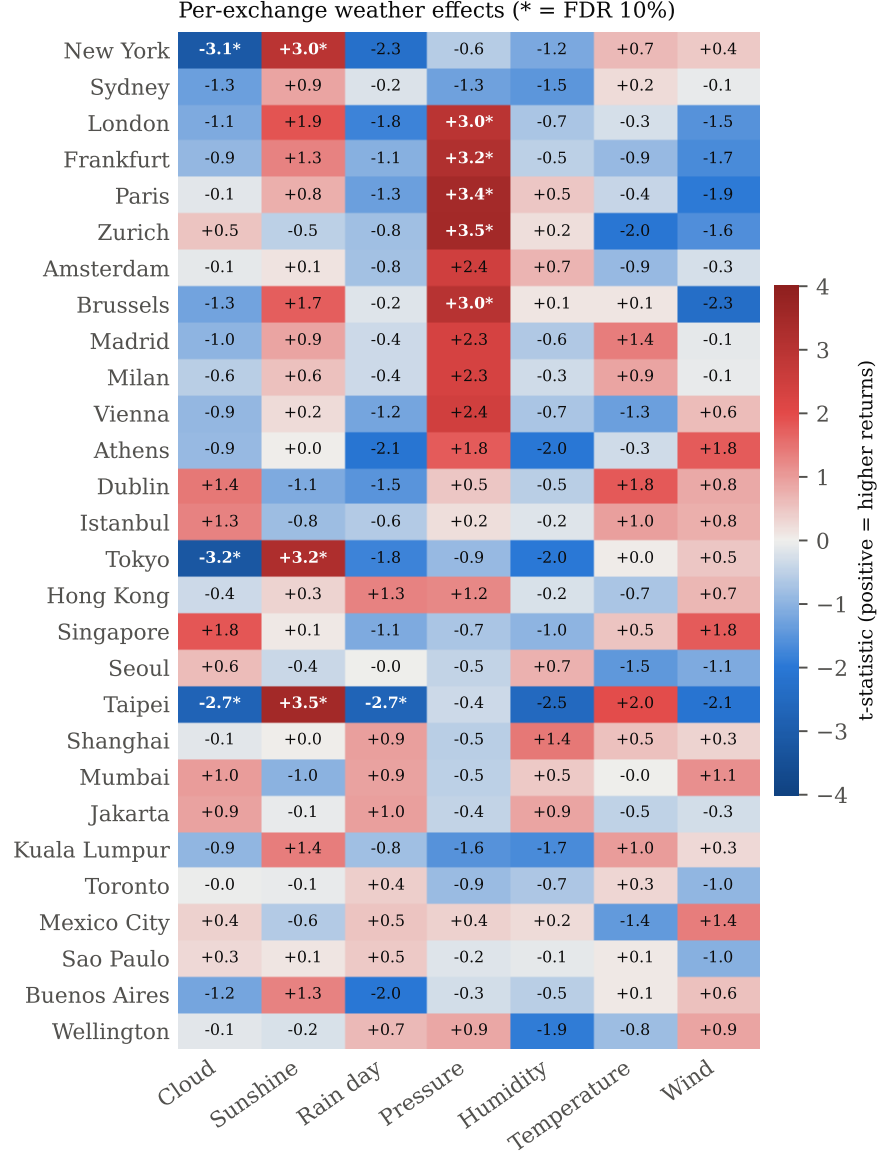


Figure 4: Exchange-level t -statistics for each weather variable (volatility-scaled returns, HAC standard errors). Positive values mean higher returns. Starred cells survive a Benjamini–Hochberg false-discovery rate of 10% across all 196 tests.

7 Discussion and conclusion

Pre-registration and placebo inference change how the old weather results should be read. The New York cloud effect is not an artefact of specification search. It replicates, survives multiplicity adjustment, and is more extreme than the same statistic computed with the weather of every one of 100 unrelated cities. But it is small, and most of the evidence for it comes from before 2007. After publication the US estimate is similar in size but imprecise, and the Australian estimate is close to zero, with a confidence interval tight enough to rule out effects of the size originally reported. The SAD cycle does not appear in Australia, and temperature matters only for industries whose cash flows depend on it. Study 2 generalises the picture. Across 26 further exchanges, fair weather (no rain, high pressure, sunshine) went with higher returns, and rain survives the full pre-registered decision rule. But these effects are concentrated before 2003, and the international cloud effect of [Hirshleifer and Shumway \(2003\)](#) does not survive out of sample.

A cautious reading is that a genuine but small mood effect existed when trading was concentrated among people who experienced the local weather, and that it has faded as trading dispersed geographically and algorithmically. The decline in the Sydney coefficient and the near-zero New York estimates in windows after 2005 fit this reading, although our direct market-structure test (H5) is too imprecise to confirm it. Future work could use investor-location data ([Goetzmann et al., 2015](#)) or intraday data around the electronic-trading transitions to sharpen the test.

References

- Bouman, S. and Jacobsen, B. (2002). The Halloween indicator, “Sell in May and go away”: Another puzzle. *American Economic Review*, 92(5):1618–1635.
- Cao, M. and Wei, J. (2005). Stock market returns: A note on temperature anomaly. *Journal of Banking & Finance*, 29(6):1559–1573.
- Goetzmann, W. N., Kim, D., Kumar, A., and Wang, Q. (2015). Weather-induced mood, institutional investors, and stock returns. *Review of Financial Studies*, 28(1):73–111.
- Hersbach, H. et al. (2020). The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146(730):1999–2049.
- Hirshleifer, D. and Shumway, T. (2003). Good day sunshine: Stock returns and the weather. *Journal of Finance*, 58(3):1009–1032.
- Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70.
- Jacobsen, B. and Marquering, W. (2008). Is it the weather? *Journal of Banking & Finance*, 32(4):526–540.
- Kamstra, M. J., Kramer, L. A., and Levi, M. D. (2003). Winter blues: A SAD stock market cycle. *American Economic Review*, 93(1):324–343.

- Keef, S. P. and Roush, M. L. (2007). Daily weather effects on the returns of Australian stock indices. *Applied Financial Economics*, 17(3):173–184.
- Kelly, P. J. and Meschke, F. (2010). Sentiment and stock returns: The SAD anomaly revisited. *Journal of Banking & Finance*, 34(6):1308–1326.
- Loughran, T. and Schultz, P. (2004). Weather, stock returns, and the impact of localized trading behavior. *Journal of Financial and Quantitative Analysis*, 39(2):343–364.
- McLean, R. D. and Pontiff, J. (2016). Does academic research destroy stock return predictability? *Journal of Finance*, 71(1):5–32.
- Newey, W. K. and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708.
- Novy-Marx, R. (2014). Predicting anomaly performance with politics, the weather, global warming, sunspots, and the stars. *Journal of Financial Economics*, 112(2):137–146.
- Politis, D. N. and Romano, J. P. (1994). The stationary bootstrap. *Journal of the American Statistical Association*, 89(428):1303–1313.
- Romano, J. P. and Wolf, M. (2005). Stepwise multiple testing as formalized data snooping. *Econometrica*, 73(4):1237–1282.
- Saunders, E. M. (1993). Stock prices and Wall Street weather. *American Economic Review*, 83(5):1337–1345.
- Trombley, M. A. (1997). Stock prices and Wall Street weather: Additional evidence. *Quarterly Journal of Business and Economics*, 36(3):11–21.
- Worthington, A. (2009). An empirical note on weather effects in the Australian stock market. *Economic Papers*, 28(2):148–154.

A Additional tables

Table 6: Robustness of the weather tests (exploratory; not multiplicity-adjusted).

Specification	Pred.	β (bp)	t	p	N
H1-US station (primary)	<0	-2.13	-2.72	0.003	17,736
H1-US ERA5 weather	<0	-1.94	-2.56	0.005	18,036
H1-US CHI weather (ERA5)	<0	-1.24	-1.61	0.054	18,036
H1-US full-sample deseasonalisation	<0	-2.07	-2.90	0.002	19,028
H1-US logit up/down	<0	-5.79	-3.73	<0.001	17,736
H1-US GARCH(1,1)-t mean	<0	-2.10	-4.24	<0.001	17,736
H1-AU station (primary)	<0	-1.77	-1.87	0.031	10,111
H1-AU ERA5 weather	<0	-1.47	-1.68	0.046	10,633
H1-AU MEL weather (ERA5)	<0	+0.24	+0.32	0.625	10,633
H1-AU full-sample deseasonalisation	<0	-2.03	-2.10	0.018	10,111
H1-AU AU price-only returns	<0	-1.76	-1.86	0.031	10,111
H1-AU logit up/down	<0	-4.11	-1.76	0.039	10,111
H1-AU GARCH(1,1)-t mean	<0	-1.05	-1.54	0.062	10,111
H2-US station (primary)	<0	-1.95	-1.18	0.119	17,736
H2-US ERA5 weather	<0	-1.40	-0.88	0.189	18,036
H2-US CHI weather (ERA5)	<0	-2.84	-1.69	0.046	18,036
H2-US full-sample deseasonalisation	<0	-1.97	-1.19	0.118	19,028
H2-US break at 2001-01 (HS working paper)	<0	-0.43	-0.27	0.395	17,736
H2-AU station (primary)	<0	+0.07	+0.06	0.523	10,111
H2-AU ERA5 weather	<0	-0.10	-0.10	0.462	10,633
H2-AU MEL weather (ERA5)	<0	-0.06	-0.06	0.477	10,633
H2-AU full-sample deseasonalisation	<0	-0.10	-0.08	0.469	10,111
H2-AU AU price-only returns	<0	+0.06	+0.05	0.521	10,111
H2-AU break at 2001-01 (HS working paper)	<0	-0.61	-0.54	0.295	10,111
H4-US station (primary)	<0	+0.13	+0.19	0.574	18,538
H4-US ERA5 weather	<0	+0.00	+0.00	0.502	18,036
H4-US CHI weather (ERA5)	<0	+0.74	+1.13	0.870	18,036
H4-US full-sample deseasonalisation	<0	-0.06	-0.09	0.463	19,938
H4-US logit up/down	<0	+2.09	+1.46	0.928	18,538
H4-US GARCH(1,1)-t mean	<0	+0.56	+1.28	0.899	18,538
H4-AU station (primary)	<0	+0.67	+0.85	0.803	10,610
H4-AU ERA5 weather	<0	+0.80	+0.98	0.837	10,633
H4-AU MEL weather (ERA5)	<0	+0.78	+1.02	0.847	10,633
H4-AU full-sample deseasonalisation	<0	+0.49	+0.60	0.725	10,610
H4-AU AU price-only returns	<0	+0.68	+0.87	0.808	10,610
H4-AU logit up/down	<0	+1.05	+0.52	0.700	10,610
H4-AU GARCH(1,1)-t mean	<0	+0.07	+0.12	0.547	10,610
RAIN-US (06-16h rain dummy)	<0	-5.42	-2.86	0.002	17,290
RAIN-AU (06-16h rain dummy)	<0	-0.00	-0.59	0.279	5,715

Table 7: Seasonal tests: SAD, FALL and the Halloween effect (exploratory).

Specification	Pred.	β (bp)	t	p	N
US SAD, 1955-2026	>0	+2.02	+2.40	0.008	18,036
US FALL, 1955-2026	<0	-2.55	-1.14	0.127	18,036
US SAD, 1928-2026	>0	+1.39	+1.85	0.032	25,866
US FALL, 1928-2026	<0	-3.01	-1.49	0.068	25,866
US SAD, post-2003	>0	+1.95	+1.09	0.139	5,829
US FALL, post-2003	<0	-3.76	-0.81	0.210	5,829
AU SAD (own season), alone	>0	+0.22	+0.20	0.419	10,633
AU SAD_NYC (calendar), alone	>0	-0.16	-0.19	0.576	10,633
AU Halloween (Nov-Apr), alone	>0	+1.38	+0.86	0.194	10,633
AU SAD with Halloween	>0	+1.55	+0.94	0.174	10,633
AU Halloween with SAD	>0	+2.87	+1.19	0.116	10,633
AU Halloween, pre-2003	>0	+1.61	+0.67	0.252	4,773
AU Halloween, post-2003	>0	+0.78	+0.38	0.353	5,860

Table 8: Market-structure and size tests (exploratory).

Specification	Pred.	β (bp)	t	p	N
US cloud x post-2020-03 floor closure	>0	-1.92	-0.58	0.718	17,736
US cloud, 1955-2006	<0	-2.05	-2.53	0.006	12,806
US cloud, 2007-2026	<0	-1.87	-0.97	0.166	4,930
AU small-minus-large ($\hat{A}XSO - \hat{A}FLI$) on cloud	<0	-1.40	-1.16	0.122	3,213

Table 9: US industry returns on weather anomalies, controlling for the same-day market return (exploratory; two-sided p).

Specification	Pred.	β (bp)	t	p	N
Industry Util (market-adjusted) on temp_z	$\pm$	-0.98	-2.25	0.025	18,538
Industry Util (market-adjusted) on cloud_z	$\pm$	+0.47	+1.02	0.309	17,736
Industry Oil (market-adjusted) on temp_z	$\pm$	-2.03	-2.87	0.004	18,538
Industry Oil (market-adjusted) on cloud_z	$\pm$	+0.76	+1.03	0.305	17,736
Industry Coal (market-adjusted) on temp_z	$\pm$	-0.17	-0.13	0.897	18,538
Industry Coal (market-adjusted) on cloud_z	$\pm$	+5.64	+3.82	<0.001	17,736
Industry Agric (market-adjusted) on temp_z	$\pm$	-0.94	-1.15	0.251	18,538
Industry Agric (market-adjusted) on cloud_z	$\pm$	+0.27	+0.30	0.768	17,736
Industry Food (market-adjusted) on temp_z	$\pm$	-0.51	-1.17	0.243	18,538
Industry Food (market-adjusted) on cloud_z	$\pm$	-0.22	-0.47	0.639	17,736
Industry Rtail (market-adjusted) on temp_z	$\pm$	+0.51	+1.19	0.233	18,538
Industry Rtail (market-adjusted) on cloud_z	$\pm$	-0.16	-0.35	0.725	17,736
Industry Meals (market-adjusted) on temp_z	$\pm$	+0.33	+0.56	0.574	18,538
Industry Meals (market-adjusted) on cloud_z	$\pm$	+0.06	+0.10	0.919	17,736
Industry Insur (market-adjusted) on temp_z	$\pm$	-0.77	-1.39	0.165	18,538
Industry Insur (market-adjusted) on cloud_z	$\pm$	-0.32	-0.54	0.589	17,736

B Deviations from the pre-registration

Study 2 had no deviations from its pre-registration. For Study 1, six deviations were logged in `prereg/deviations.md`; the table there is authoritative. D1–D5 were made before the corresponding results were seen. D1: weather sources, because NOAA discontinued the ISD-Lite files in 2025. D2: at least three valid hourly reports per day instead of four, because Sydney reported three-hourly before 2000. D3: dividend add-back source, because the Reserve Bank of Australia table named in the plan no longer exists. D4: H3 judged on the Holm test alone, because no placebo can be defined for an astronomical variable. D5: ERA5 obtained from the Copernicus Climate Data Store rather than a rate-limited third-party API. D6 is a bug fix made after results were seen: the year-shifted placebo had dropped non-matching weekdays. It affects only that secondary statistic; both versions are archived.